OBD2 ECU Simulator

User Manual



IMSB5010

http://www.imsapp.com

Modes

MODE 01:Show current data

MODE 02:Show freeze frame data

MODE 03:Show stored Diagnostic Trouble Codes

MODE 04:Clear Diagnostic Trouble Codes and stored values

MODE 05:Test results, oxygen sensor monitoring (non CAN only)

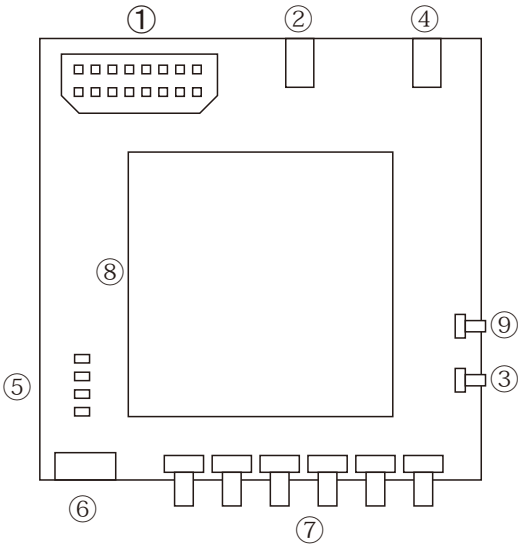
MODE 06:Test results, other component/system monitoring
(Test results, oxygen sensor monitoring for CAN only)

MODE 07:Show pending Diagnostic Trouble Codes
(detected during current or last driving cycle)

MODE 09:Request vehicle information

MODE 0A:Permanent Diagnostic Trouble Codes (DTCs)
(Cleared DTCs)

Diagram



①Diagnostic Link Connector (DLC)

②Power jack (13 VDC)

③MIL button

④USB connector

⑤Status LEDs

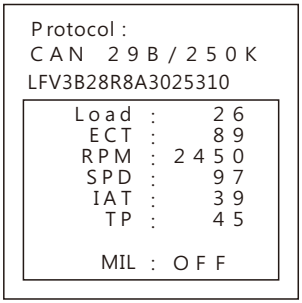
⑥Protocol selection

⑦Six knobs (set PID value)

⑧Liquid Crystal Display

⑨Reset button



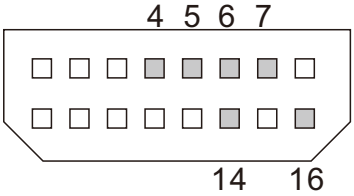


←Protocol

←VIN

←MIL status

OBD2 Socket



4: Chassis ground
5: Signal ground
6: CAN High
7: ISO-K LINE
14: CAN Low
16: Battery power

DC IN

Input : 100~240VAC

Output : 13VDC - 1A -⊂+

MIL button

When pressed, the following happens:

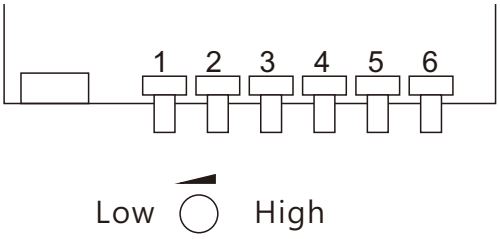
1. Set MIL and generate DTCs
2. Generate pending, stored, and permanent DTCs
3. Generate freeze frame data
4. MIL led ON

Status LEDs

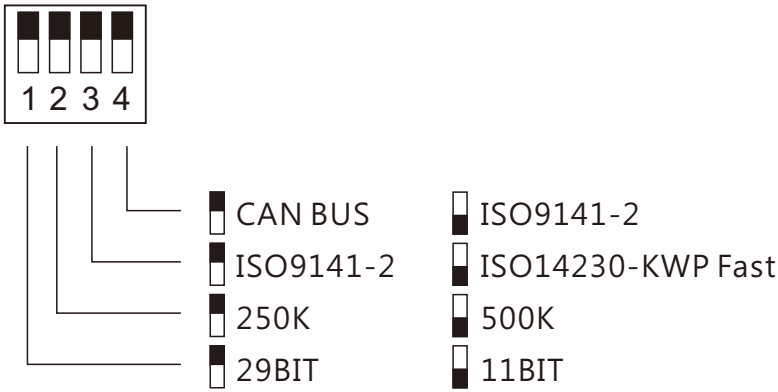
CHECK：flashing=simulator is functioning properly.
DATA：OBD message is received
LINK：ECU is initialized.
MIL：when MIL button pressed.
(turn off automatic after clear DTC command received)

Knobs

Six knobs assigned to the following commonly used Mode 1 PIDs.
1. PID 10: MAF air flow rate
2. PID 05: Temperature of the engine coolant in °C
3. PID 0C: Engine speed in rpm
4. PID 0D: Vehicle speed in kph
5. PID 0F: Intake air temperature in °C
6. PID 11: Throttle position in %



Protocol Selection





ISO15765-4 11BIT/500K



ISO15765-4 29BIT/500K



ISO15765-4 11BIT/250K



ISO15765-4 29BIT/250K



ISO9141-2



ISO14230-KWP Fast

Comm Port

USB Port

USB Chip: CP2102

The communication settings are:

Baud rate: 115200 bps

Data bits: 8

Parity: none

Stop bits: 1

https://www.silabs.com

Data Format

<R>: <E>: [18DB33F1] 01 3C

<T>: <E>: [18DAF110] 41 3C 11 90

} ISO15765-4 CAN 29Bit

<R>: <S>: [000007DF] 01 10

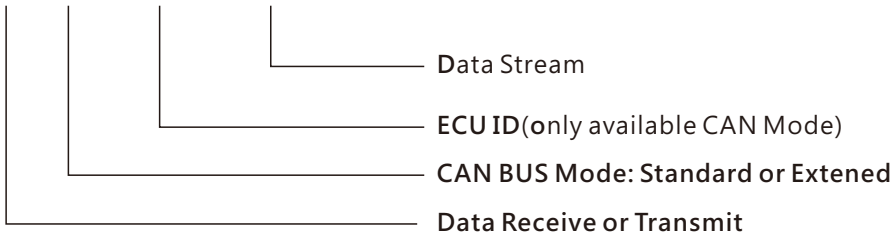
<T>: <S>: [000007E8] 41 10 01 3C

} ISO15765-4 CAN 11Bit

<R>: C1 33 F1 03 E8

<T>: 87 F1 11 43 C1 01 81 02 40 32 83

} ISO9141-2/14230



MODE 03

CAN-BUS: 29bit-Extened

P0501-Vehicle Speed Sensor 'A' Range/Performance

P0336-Crankshaft Position Sensor A Circuit Range/Performance

CAN-BUS: 11bit-Standard

P0168-Fuel Temperature Too High

P0303-Cylinder 3 Misfire Detected

ISO9141-2、ISO14230-KWP Fast:

U0101-TCM Lost Communication

B0102-undocumented

C0032-Left front wheel speed sensor supply

Pending DTCs: MODE 07

Permanent DTCs: MODE 0A

P0103-Mass or Volume Air Flow Circuit High Input

P0104-Mass or Volume Air Flow A Circuit Intermittent

Freeze Frame

***Freeze frame data:**

1. DTC P0168 or P0001
2. Engine Load 12.5%
3. Engine coolant temperature 99°C
4. Intake manifold absolute pressure 32kPa
5. Engine RPM 500rpm
6. Vehicle speed 0km/h
7. MAF air flow rate 3.15g/s
8. Throttle position 12.9%

Vehicle Data

Vehicle Identification Number (VIN): LFV3B28R8A3025310
Calibration Identifications (CALID): 8R0907115A Q0006
Calibration Verification Numbers (CVN): 0A19B32B
ECU Name: ECM

PID List

□fixed ▭variable ■knob

PID	Description	Variable/Fixed	Min	Max
00	List of PIDs supported (1 to 1F)	fixed		
01	Monitor status since DTCs cleared.	fixed		
02	Freeze DTC	fixed		
03	Fuel system status	variable	Closed	
04	Engine load calculated in %	variable	0	100
05	Temperature of the engine coolant in Centigrade	Knob 2	-40	215
06	Short-term fuel % trim bank 1	variable	-100	99.22
07	Long-term fuel % trim bank 1	variable	-100	99.22
08	Short-term fuel % trim bank 2	variable	-100	99.22
09	Long-term fuel % trim bank 2	variable	-100	99.22
0A	Fuel pressure in kPa (gauge)	variable	0	765
0B	Intake manifold absolute pressure in kPa (absolute)	variable	0	255
0C	Engine speed in rpm	Knob 3	0	16,383.00
0D	Vehicle speed in kph	Knob 4	0	255
0E	Timing advance on cylinder 1 in degrees	variable	-64	63.5
0F	Intake air temperature in Centigrade	Knob 5	-40	215
10	Air flow measured by the flowmeter in g/s	variable	0	655.35
11	Throttle position in %	Knob 6	0	100
12	Status of the secondary intake circuit	fixed	air reserved	
13	O2 sensor positions bank/sensor	fixed		
14	Oxygen sensor volts bank 1 sensor 1	fixed	0.080v	0%
15	Oxygen sensor volts bank 1 sensor 2	fixed	0.160v	12.50%
16	Oxygen sensor volts bank 1 sensor 3	fixed	0.240v	25.00%
17	Oxygen sensor volts bank 1 sensor 4	fixed	0.320v	37.50%
18	Oxygen sensor volts bank 2 sensor 1	fixed	0.400v	50.00%
19	Oxygen sensor volts bank 2 sensor 2	fixed	0480v	-12.50%
1A	Oxygen sensor volts bank 2 sensor 3	fixed	0.560v	-25.00%
1B	Oxygen sensor volts bank 2 sensor 4	fixed	0.640v	-37.50%
1C	OBD computer specification	fixed		
1D	O2 sensor positions bank/sensor	fixed		
1E	Auxiliary input status	fixed	ON	
1F	Run time since engine start	variable	0	65,535
20	List of PIDs supported (21 to 3F)	fixed		
21	Distance travelled with MIL on in kms	variable	0	65,535
22	Relative fuel rail pressure in kPa	variable	0	5177.265
23	Fuel rail pressure in kPa (gauge)	variable	0	655,350
24	O2 sensor (extended range) bank 1, sensor 1 (lambda and volts)	fixed	0.125	3.998v
25	O2 sensor (extended range) bank 1, sensor 2 (lambda and volts)	fixed	0.25	3.498v
26	O2 sensor (extended range) bank 1, sensor 3 (lambda and volts)	fixed	0.375	2.998v
27	O2 sensor (extended range) bank 1, sensor 4 (lambda and volts)	fixed	0.5	2.499v
28	O2 sensor (extended range) bank 2, sensor 1 (lambda and volts)	fixed	0.625	1.999v
29	O2 sensor (extended range) bank 2, sensor 2 (lambda and volts)	fixed	0.75	1.499v
2A	O2 sensor (extended range) bank 2, sensor 3 (lambda and volts)	fixed	0.874	0.999v
2B	O2 sensor (extended range) bank 2, sensor 4 (lambda and volts)	fixed	0.999	0.500v
2C	EGR in %	variable	0	100
2D	EGR error in %	variable	-100	99.22
2E	Evaporation purge in %	variable	0	100
2F	Fuel level in %	variable	0	100

30	Number of warning(s) since faults (DTC) were cleared	variable	0	255
31	Distance since faults (DTC) were cleared.	variable	0	65,535
32	Evaporation system vapour pressure in Pa	variable	-8,192	8,192
33	Barometic pressure in kPa (Absolute)	variable	0	255
34	O2 sensor (extended range) bank 1, sensor 1 (lambda and volts)	fixed	1.001	0.012
35	O2 sensor (extended range) bank 1, sensor 2 (lambda and volts)	fixed	0.25	16
36	O2 sensor (extended range) bank 1, sensor 3 (lambda and volts)	fixed	0.375	32
37	O2 sensor (extended range) bank 1, sensor 4 (lambda and volts)	fixed	0.5	48
38	O2 sensor (extended range) bank 2, sensor 1 (lambda and volts)	fixed	0.625	64
39	O2 sensor (extended range) bank 2, sensor 2 (lambda and volts)	fixed	0.75	80
3A	O2 sensor (extended range) bank 2, sensor 3 (lambda and volts)	fixed	0.874	96
3B	O2 sensor (extended range) bank 2, sensor 4 (lambda and volts)	fixed	0.999	112
3C	Catalyst temperature in Centigrade bank 1, sensor 1	variable	-40	6,513.50
3D	Catalyst temperature in Centigrade bank 2, sensor 1	variable	-40	6,513.50
3E	Catalyst temperature in Centigrade bank 1, sensor 2	variable	-40	6,513.50
3F	Catalyst temperature in Centigrade bank 2, sensor 1	variable	-40	6,513.50
40	List of PIDs supported (41 to 5F)	fixed		
41	Monitor status this drive cycle	fixed		
42	Control module voltage in V	variable	0	65.535
43	Absolute engine load	variable	0	25,700
44	Equivalent fuel/air mixture request	variable	0	2
45	Relative throttle position in %	fixed	2.70%	
46	Ambient air temperature in Centigrade	variable	-40	215
47	Absolute throttle position B in %	variable	0	100
48	Absolute throttle position C in %	variable	0	100
49	Accelerator pedal position D in %	variable	0	100
4A	Accelerator pedal position E in %	variable	0	100
4B	Accelerator pedal position F in %	variable	0	100
4C	Commanded throttle actuator in %	variable	0	100
4D	Engine run time since MIL on in min	variable	0	65,535
4E	Engine run time since faults cleared in min	variable	0	65,535
4F	Exteral test equipment no. 1 configuration information	fixed		
50	Exteral test equipment no. 2 configuration information	fixed		
51	Fuel type used by the vehicle	variable		
52	Ethanol fuel %	variable	0	100
53	Absolute evaporation system vapour pressure in kPa	variable	0	327.675
54	Evaporation system vapour pressure in Pa	Knob 1	-32,767	32,768
55	Short-term O2 sensor trim bank 1 and 3	fixed	-87.50%	
56	Long-term O2 sensor trim bank 1 and 3	fixed	-0.80%	
57	Short-term O2 sensor trim bank 2 and 4	fixed	-87.50%	
58	Long-term O2 sensor trim bank 2 and 4	fixed	-87.50%	
59	Absolute fuel rail pressure in kPa	variable	0	655,350
5A	Relative accelerator pedal position in %	variable	0	100
5B	Battery unit remaining life (hybrid) in %	variable	0	100
5C	Engine oil temperature in Centigrade	variable	-40	210
5D	Fuel injection timing in ?	variable	-210	301.992
5E	Fuel consumption in litre/hr	variable	0	3212.75
5F	Fuel consumption in litre/hr	fixed		

Setting the VIN:

$VIN[VALUE]

VIN must be 17 ASCII characters.

example:

$VINLFV3B28R8A3025310|

Setting the CID:

$CID[VALUE]

CVN 3 to 16 ASCII characters.

example:

$CID8R0907115A-Q0006|

Setting the CVN:

$CVN[VALUE]

CVN must be 8 HEX characters.

example:

$CVN0A19B32B|

Setting the ECU Name:

$ECU[ECU NAME]

ECU 2 to 16 ASCII characters.

example:

$ECUENGINE_MODULE|

return:

OK> command successfully executed

ERR> instruction / value was incorrect or communication error.

Command

Setting the PID value:

$PID[PID]:[VALUE]
[PID] 2 hexadecimal digits
[VALUE] 1 to 6 decimal digits

example:
set Run time since engine start (1234 seconds)
$PID1F:1234|

set Distance since faults (DTC) were cleared. (125km)
$PID31:125|

set Catalyst temperature in °C bank 1, sensor 1 (-12°C)
$PID3C:-12|

MODE 1 PID01 Readiness Monitors

$SSS[HHHHHH]
3 Hex digitsis.

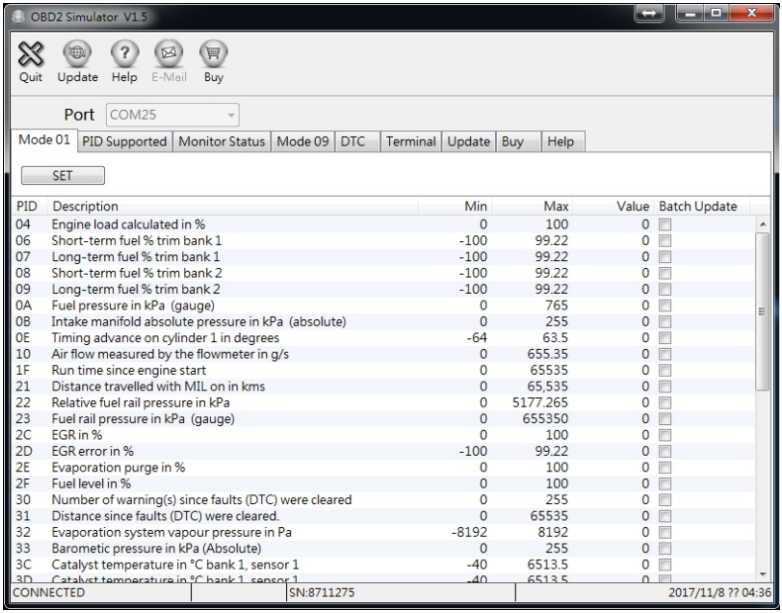
example:
$SSS07FF00|

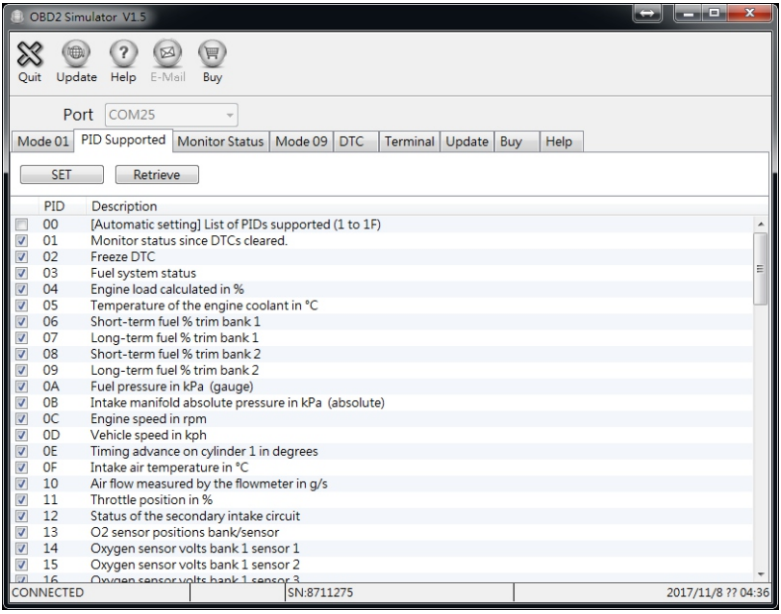
Software

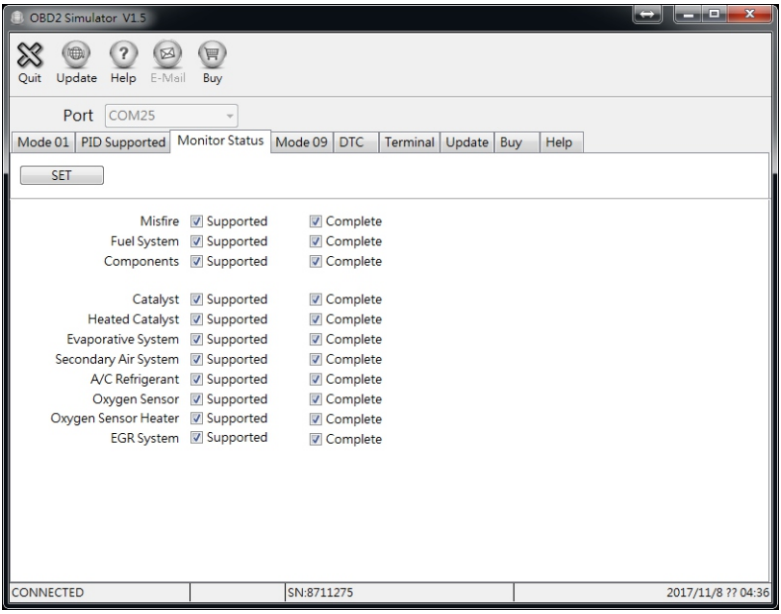
System require:

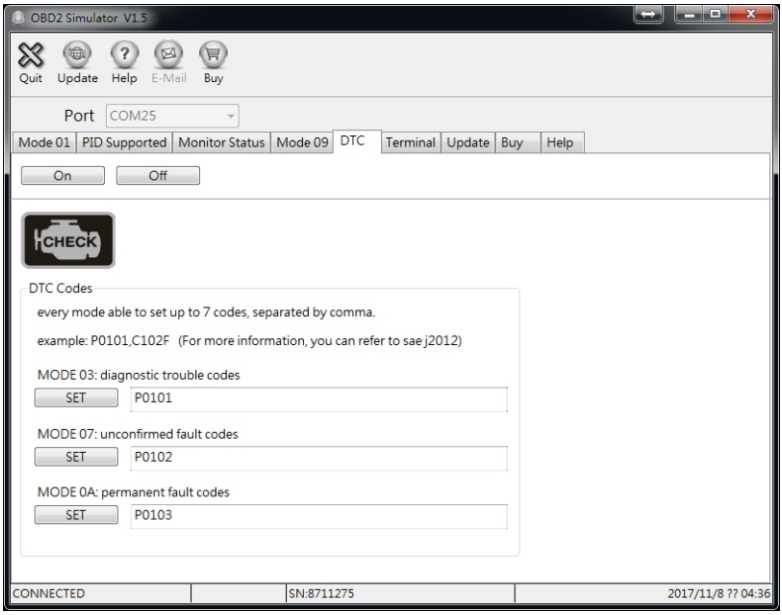
Windows XP Sp3 / 7 / 8 / 10

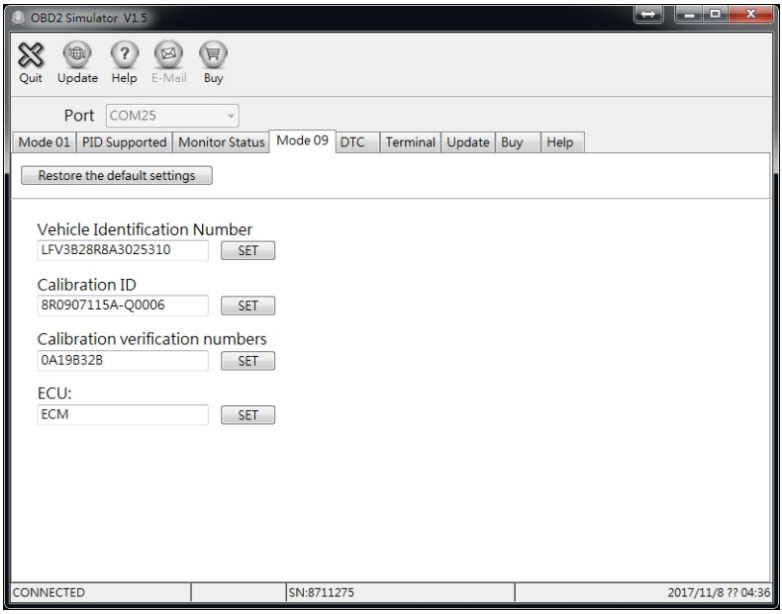
Mac OS 10.9




OBD2 Simulator V1.5
Quit
Update
Help
E-Mail
Buy
Port
COM25
Mode 01
PID Supported
Monitor Status
Mode 09
DTC
Terminal
Update
Buy
Help
SET
Retrieve
PID
Description
00 [Automatic setting] List of PIDs supported (1 to 1F)
01 Monitor status since DTCs cleared.
02 Freeze DTC
03 Fuel system status
04 Engine load calculated in %
05 Temperature of the engine coolant in °C
06 Short-term fuel % trim bank 1
07 Long-term fuel % trim bank 1
08 Short-term fuel % trim bank 2
09 Long-term fuel % trim bank 2
0A Fuel pressure in kPa (gauge)
0B Intake manifold absolute pressure in kPa (absolute)
0C Engine speed in rpm
0D Vehicle speed in kph
0E Timing advance on cylinder 1 in degrees
0F Intake air temperature in °C
10 Air flow measured by the flowmeter in g/s
11 Throttle position in %
12 Status of the secondary intake circuit
13 O2 sensor positions bank/sensor
14 Oxygen sensor volts bank 1 sensor 1
15 Oxygen sensor volts bank 1 sensor 2
CONNECTED
SN:8711275
2017/11/8 ?? 04:36


OBD2 Simulator V1.5
Quit
Update
Help
E-Mail
Buy
Port
COM25
Mode 01
PID Supported
Monitor Status
Mode 09
DTC
Terminal
Update
Buy
Help
SET
Misfire Supported Complete
Fuel System Supported Complete
Components Supported Complete
Catalyst Supported Complete
Heated Catalyst Supported Complete
Evaporative System Supported Complete
Secondary Air System Supported Complete
A/C Refrigerant Supported Complete
Oxygen Sensor Supported Complete
Oxygen Sensor Heater Supported Complete
EGR System Supported Complete
CONNECTED
SN:8711275
2017/11/8 ?? 04:36


OBD2 Simulator V1.5
Quit
Update
Help
E-Mail
Buy
Port
COM25
Mode 01
PID Supported
Monitor Status
Mode 09
DTC
Terminal
Update
Buy
Help
On
Off
CHECK
DTC Codes
every mode able to set up to 7 codes, separated by comma.
example: P0101,C102F (For more information, you can refer to sae j2012)
MODE 03: diagnostic trouble codes
SET
P0101
MODE 07: unconfirmed fault codes
SET
P0102
MODE 0A: permanent fault codes
SET
P0103
CONNECTED
SN:8711275
2017/11/8 ?? 04:36


OBD2 Simulator V1.5
Quit
Update
Help
E-Mail
Buy
Port
COM25
Mode 01
PID Supported
Monitor Status
Mode 09
DTC
Terminal
Update
Buy
Help
Restore the default settings
Vehicle Identification Number
LFV3B28R8A3025310
SET
Calibration ID
8R0907115A-Q0006
SET
Calibration verification numbers
0A19B32B
SET
ECU:
ECM
SET
CONNECTED
SN:8711275
2017/11/8 ?? 04:36